MODULE 01

Python for Data Science

A concise 10-page starter workbook for Data Science students

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Learning goal

This workbook introduces the core ideas in python for data science, with a topic-by-topic study path, practical checkpoints, and reflection prompts. Use it as a starting point and supplement it with your class notes and recommended references.

Note: These are sample learning materials created for this demo portal, not official university course notes.

Variables, Data Types, and Operators

Concept overview

In this section, focus on the key principles behind **variables, data types, and operators**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain variables, data types, and operators to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Control Flow and Iteration

Concept overview

In this section, focus on the key principles behind **control flow and iteration**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain control flow and iteration to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Functions and Reusable Code

Concept overview

In this section, focus on the key principles behind **functions and reusable code**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain functions and reusable code to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Lists, Tuples, Sets, and Dictionaries

Concept overview

In this section, focus on the key principles behind **lists, tuples, sets, and dictionaries**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain lists, tuples, sets, and dictionaries to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

NumPy Arrays and Vectorized Operations

Concept overview

In this section, focus on the key principles behind **numpy arrays and vectorized operations**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain numpy arrays and vectorized operations to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Pandas Series and DataFrames

Concept overview

In this section, focus on the key principles behind **pandas series and dataframes**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain pandas series and dataframes to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Data Cleaning with Pandas

Concept overview

In this section, focus on the key principles behind **data cleaning with pandas**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain data cleaning with pandas to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Exploratory Data Analysis

Concept overview

In this section, focus on the key principles behind **exploratory data analysis**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain exploratory data analysis to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Mini Project: Student Performance Analysis

Concept overview

In this section, focus on the key principles behind **mini project: student performance analysis**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain mini project: student performance analysis to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?